Question:

A group of scientists is conducting an experiment on the human cells involved in the immune response. They genetically modify B cells so they do not express the cluster of differentiation 21 (CD21) on their cell surfaces. The pathogenesis of which of the following organisms would most likely be affected by this genetic modification?

Answer Choices:

A. Epstein-Barr virus (EBV)

B. Measles virus

C. Human immunodeficiency virus (HIV)

D. Human papillomavirus

Reasoning + Answer:

<thought>
To solve this problem , we need to understand what CD 2 1 does . CD 2 1 is a complement receptor type 1 (CR 1) that participates in the complement system , which is involved in the immune response .
</thought>
<answer>
A . Epstein -Barr virus (EB V)
</answer>
<thought>
The EB V virus interacts with CD 2 1 . The human immune system recognizes EB V -infected cells via interactions that involve CD 2 1 on the B cells . By genetically modifying B cells to no longer express CD 2 1 , the interaction with EB V -infected cells would be impaired , potentially leading to a pathogenesis issue . However , the other viruses (Measles virus , HIV , and Human papillomavirus) do not interact with CD 2 1 in the same way that EB V does , as these viruses use other pathways for their infection cycles .</thought>
<answer>
A . Epstein -Barr virus (EB V)
</answer>